Haesung Pyun

Ph.D. Student in Data Science | Language Model Behavior & Reliability

haesung.pyun@snu.ac.kr | Website | GitHub | Google Scholar Seoul, South Korea

RESEARCH FOCUS

I study why information available within language models does not always guide their predictions, and how to diagnose and reduce this gap. I am interested in extending this work toward reliable evaluation, uncertainty-aware prediction, and sequential decision-making for adaptive AI systems.

EDUCATION

Seoul National University Sep 2023 - Aug 2028 (expected)

Ph.D. in Data Science, Graduate School of Data Science | Advisor: Yohan Jo

Hanyang University 2017 - 2023

B.S. in Electronic Engineering, Magna Cum Laude

RESEARCH EXPERIENCE

Seoul National University | Ph.D. Researcher 2023 - Present

- KAPPA (ICML 2026, co-first author). Analyzed failures where correct-answer signals are linearly decodable from LLM hidden representations but predictions are incorrect; characterized the gap via residual-stream probes and geometry.
- Co-developed a parameter-free inference-time intervention to align model predictions with latent knowledge; improved four-choice TruthfulQA accuracy from 56.7% to 73.5% on Llama 3.1 8B and tested generalization to free-form answers.
- CombiSearch (ACL 2025, first author). Designed combinatorial in-context example selection for dialogue state tracking; achieved 20x training-data efficiency and a 12-percentage-point improvement in oracle upper-bound DST performance.
- PA-Tool (ACL 2026, co-author). Investigated model/tool-schema mismatch and contributed SFT-based validation of a training-free approach that adapts tool schemas to pretrained model knowledge.

SNU | Structure & Knowledge Injection into ML Lab Jan - Aug 2023

- SEAL-SSL. Explored energy-based objectives for semi-supervised learning; implemented knowledge distillation, REINFORCE, and task adaptation.

SNU | Machine Intelligence Laboratory Mar - Dec 2022

- Developed multi-view intent induction with clustering and multitask learning; contributed to DSTC11 Workshop research.

KIST | AI Research Center Dec 2021 - Feb 2022

- Implemented recommender models and evaluated a tourism recommendation prototype through A/B testing and user feedback.

SELECTED PUBLICATIONS

- Y. Park*, H. Pyun*, Y. Jo. Bridging the Knowledge-Prediction Gap in LLMs on Multiple-Choice Questions. ICML 2026. (* Equal contribution)
- H. Pyun, Y. Park, Y. Jo. Improving Dialogue State Tracking through Combinatorial Search for In-Context Examples. ACL 2025.
- J. Lee, W. Song, J. Han, H. Pyun, Y. Jo. Don't Adapt Small Language Models for Tools; Adapt Tool Schemas to the Models. ACL 2026.
- J. Song, S. Song, M. Oh, H. Pyun, S. Lee, Y. Jo. SHAPE of Chain-of-Thought in Math Reasoning. ICML 2026 Workshop.

HONORS, TEACHING & SERVICE

NRF Graduate Research Fellowship (2-year award) | Teaching: Computing for Data Science 1 (2026), Conversational AI (2024) | Service: Reviewer, Pluralistic Alignment Workshop @ ICML 2026.